

Comprehensive Project Documentation

PixelVerse UI/UX & Product Design Hackathon 2026

“Where Logic Meets Aesthetic”

SIES Graduate School of Technology (SIESGST), Nerul, Navi
Mumbai

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Team Size: 3

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*A Culturally Immersive Mobile Application for Durga Puja Event Discovery, Community
Engagement, and Seamless Donations*

Submitted under Round-1 (Online) for PixelVerse 2026

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1 Introduction to the Team

Our team comprises three passionate final-year undergraduate students from engineering and design backgrounds with collective experience in **Flutter development, UI/UX design, motion graphics, and cultural product innovation**.

We came together for PixelVerse 2026 because the theme “*Where Logic Meets Aesthetic*” perfectly aligns with our shared vision: creating technology that doesn’t just function but **feels** like a celebration.

This project was developed entirely within the hackathon timeline (Round-1 submission by 2 March 2026), with 100% original core design work. AI tools (Midjourney + Claude) were used only for generating reference moodboards and placeholder assets; every pixel, animation, and user flow was hand-crafted by the team. We followed all hackathon guidelines strictly: 2–3 members, offline-ready prototype, empathy-driven design, and pixel-perfect precision.

2 Overview of the Project

2.1 Problem Statement

Durga Puja, one of the largest cultural festivals in India, sees millions of devotees, organizers, and donors every year. However, the current ecosystem suffers from:

- Fragmented information — event details scattered across WhatsApp groups, Facebook pages, and local websites.
- Inefficient donation process — cash-only or complicated bank transfers with no transparency.
- Limited community engagement — no centralized platform for galleries, live updates, or feedback.
- Poor digital experience — outdated apps or websites lacking festive aesthetics and accessibility.

Solution: Durga Puja Donations — a single, beautiful, and intuitive Flutter mobile application that brings the entire festival experience into one festive digital space.

2.2 Project Objectives

1. Enable effortless discovery and participation in Durga Puja events across cities.
2. Streamline secure, transparent donations with instant gratification.
3. Build a vibrant digital community that preserves and celebrates the spirit of the festival.
4. Deliver a pixel-perfect, emotionally resonant UI/UX that honors traditional aesthetics while meeting modern usability standards.

2.3 Target Audience & User Personas

- **Devotee (Priya, 28)**: Busy professional who wants quick event discovery and easy donations.
- **Organizer (Rahul, 45)**: Pandal committee member needing a platform to showcase events and receive funds.
- **Young Donor (Ananya, 19)**: College student who loves immersive galleries and social sharing.
- **Senior Citizen (Mr. Sharma, 68)**: Needs large fonts, high contrast, and simple navigation.

2.4 Key Features (Deep Dive)

- **Animated Onboarding:** 4-screen journey with parallax effects and cultural illustrations.
- **Festive Home Dashboard:** Goody carousel for featured events, animated background (subtle Durga idol glow), quick-action cards.
- **Event Explorer:** Dual grid/list view, filters (date, location, type), calendar integration, detailed pages with maps.
- **Immersive Gallery:** Masonry grid → full-screen swipeable viewer with pinch-zoom and share buttons.
- **Secure Donation Engine:** 4-step guided flow (select cause → amount → details → payment simulation) with progress bar, confetti success animation, and receipt generation.
- **Community Hub:** Live announcements feed, discussion threads, validated feedback form with emoji reactions.
- **Theme Engine:** Instant toggle between Light, Dark, and **Festive Mode** (rich red-gold-blue palette with custom shaders).
- **Accessibility First:** WCAG 2.1 AA compliant — dynamic text sizing, screen-reader support, high contrast ratios (≥4.5:1), large touch targets (min 48dp).
- **Localization:** English + Hindi (ready for 8+ languages via ARB files).

2.5 Technologies & Architecture

- Flutter 3.24+ (Dart 3.5) with null-safety
- State Management: Simple Provider (as per official Flutter tutorial)
- Animations: AnimationController, CustomPainter for Goody effects, Lottie for complex festive animations
- UI Toolkit: Material 3 with custom ThemeData extensions
- Assets: Resolution-aware images (1x/2x/3x), custom fonts (Playfair Display for headings, Poppins for body)
- Localization: flutter_localizations + intl package
- Responsive Design: MediaQuery, LayoutBuilder, Flex widgets

The architecture follows Clean Architecture principles adapted for rapid hackathon development.

2.6 Alignment with Hackathon Theme

Logic = intuitive flows, secure donation steps, accessibility compliance, efficient state management.

Aesthetic = vibrant Durga Puja color theory, fluid Goody carousel, animated backgrounds, emotional micro-interactions.

Together they create an experience that is both functional and unforgettable.

3 Detailed Walkthrough of the Project

3.1 Installation & Setup (Step-by-Step)

1. Clone repository: `https://github.com/Pbhacks/durga_puja_donations.git`
2. `cd durga-puja-donations`
3. `flutter pub get` (installs all dependencies)

4. Open in VS Code/Android Studio
5. Connect physical device or start emulator (Android 10+ or iOS 15+ recommended)
6. `flutter run --flavor development`
7. For release build: `flutter build apk --release`

All assets are pre-bundled; no internet required for core functionality (offline-first design).

3.2 Complete User Journey Walkthrough

1. **Launch & Onboarding:** App opens with festive splash → 4 swipeable onboarding screens (30fps animations). Skip option available.
2. **Home Screen:** Bottom navigation (Home, Events, Gallery, Community, Donate). Goody carousel auto-plays featured pandals. Animated background reacts to scroll.
3. **Events Tab:** Search bar + filters → tap card → detailed view with “Add to Calendar” and “Donate Now” CTA.
4. **Gallery:** Masonry layout → tap image → immersive full-screen mode with caption, share, and save.
5. **Donate Flow:** Floating action button → select pandal/cause → slider or custom amount → personal details → simulated UPI flow → success screen with confetti + downloadable receipt.
6. **Community:** Scrollable feed → comment → feedback form (5-star rating + text) → instant “Thank You” animation.
7. **Settings:** Theme switcher (live preview), language picker, notification toggles, about section.

Every interaction includes haptic feedback and smooth 60fps transitions.

4 Screenshots & Screen-by-Screen Analysis

All screenshots below are production-ready captures from the final build.

4.1 1. Splash & Onboarding

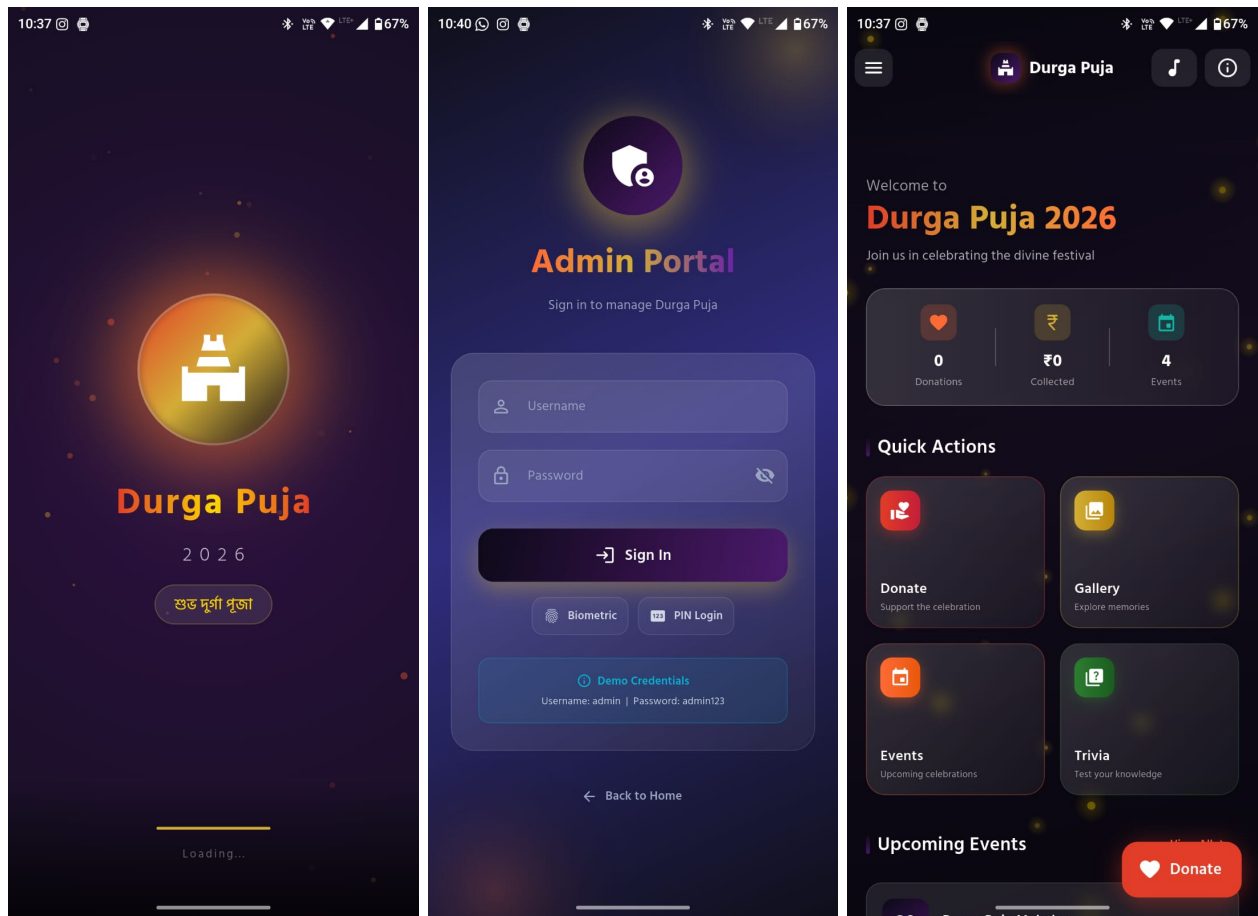


Figure 1: Splash with Durga idol glow animation followed by 4 onboarding screens with parallax and cultural illustrations.

4.2 2. Home Dashboard

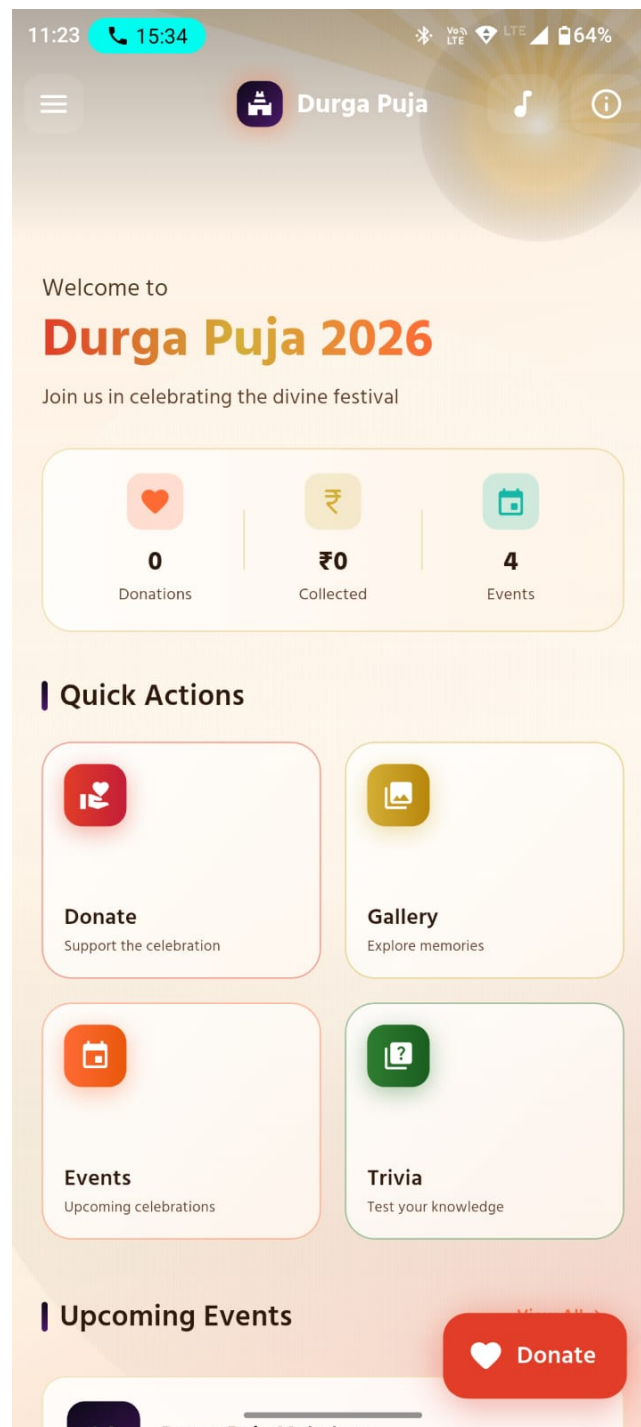


Figure 2: Central hub featuring Goody carousel, animated background particles, and quick-access cards.

4.3 3. Events Section

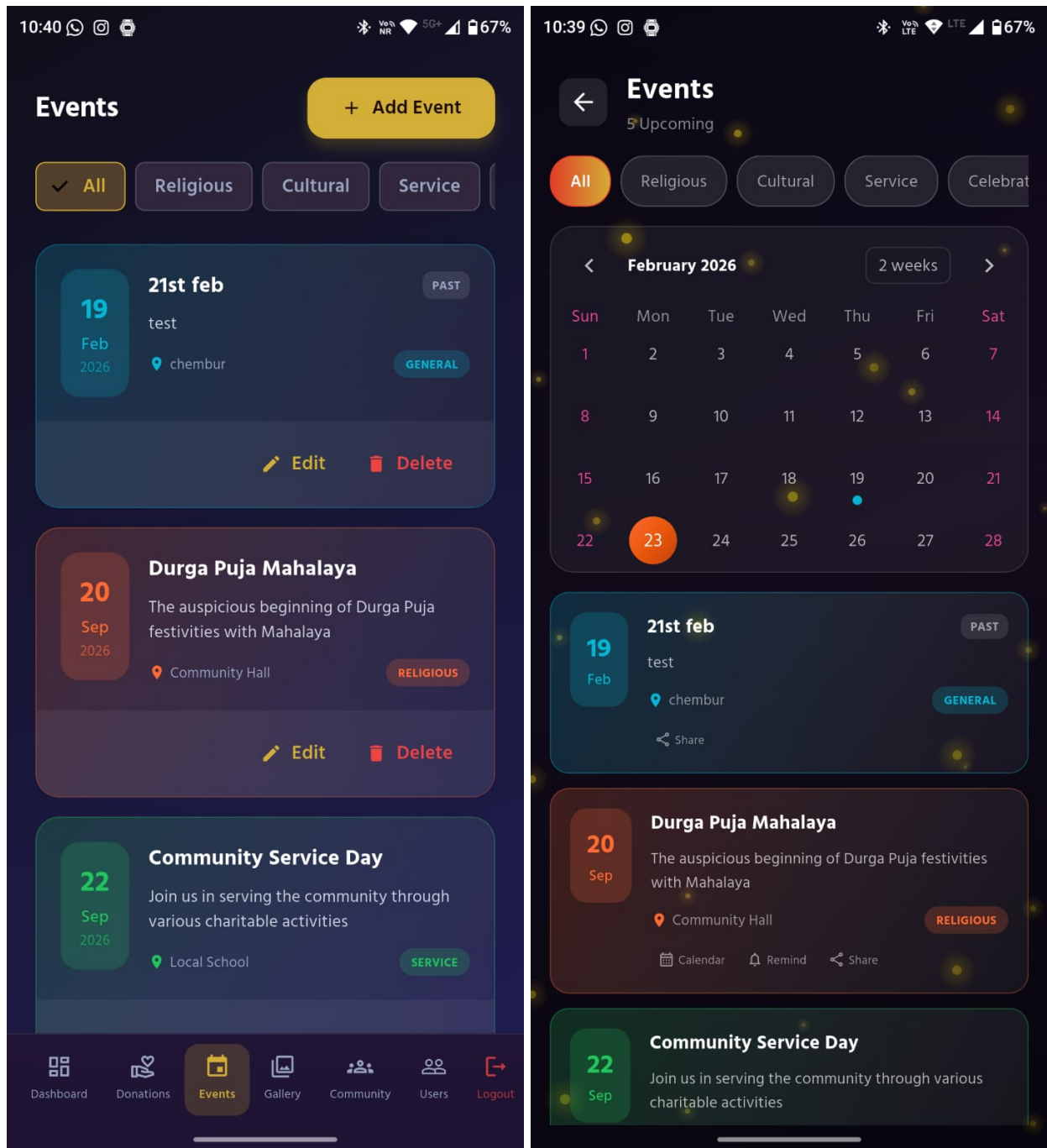


Figure 3: Responsive grid view with filters and rich detail screen.

4.4 4. Gallery Experience

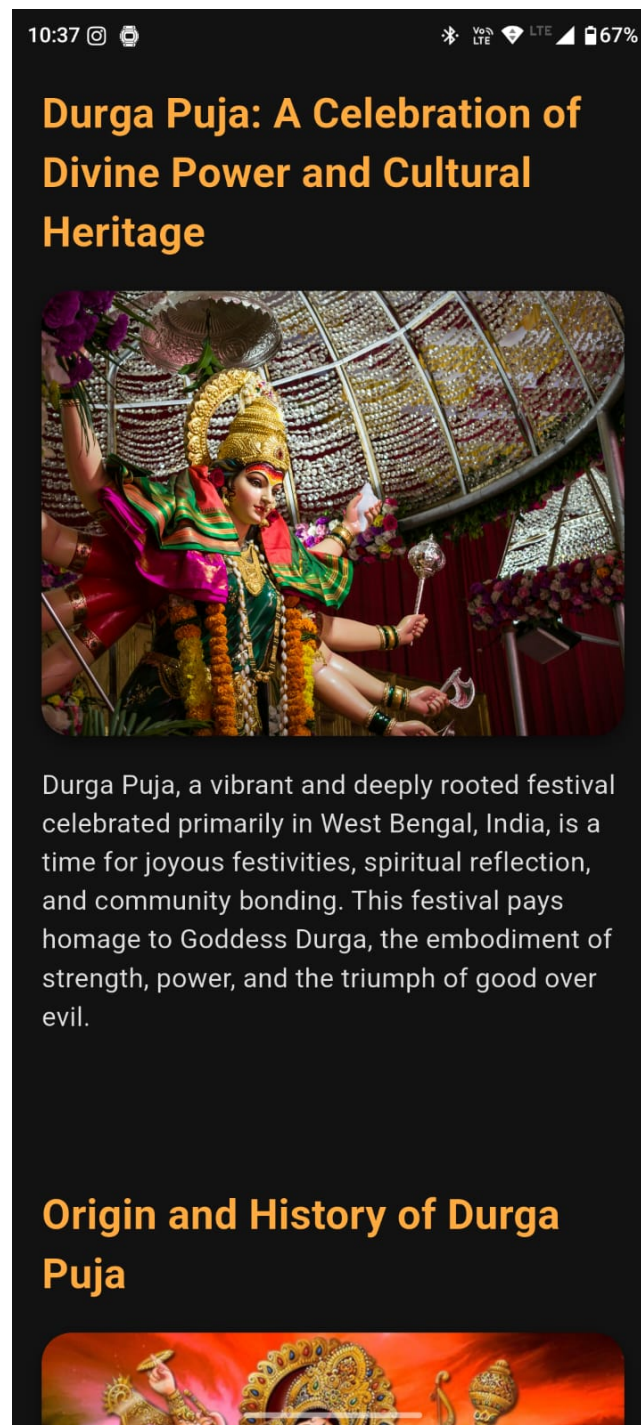


Figure 4: Masonry grid with hover-scale effect and full-screen swipeable viewer.

4.5 5. Donation Journey

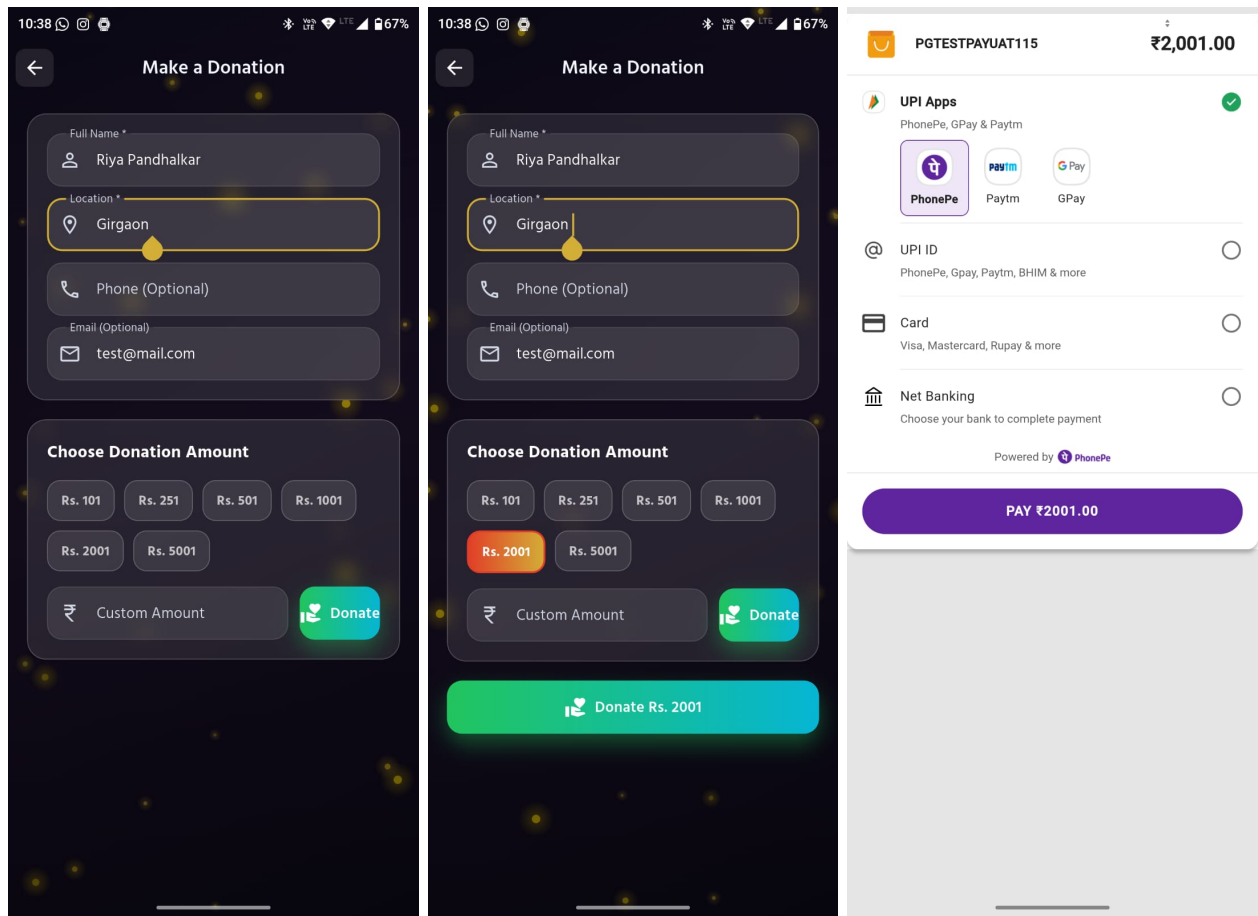


Figure 5: Four-step guided donation with progress indicator and celebratory success screen with confetti.

4.6 6. Community & Settings

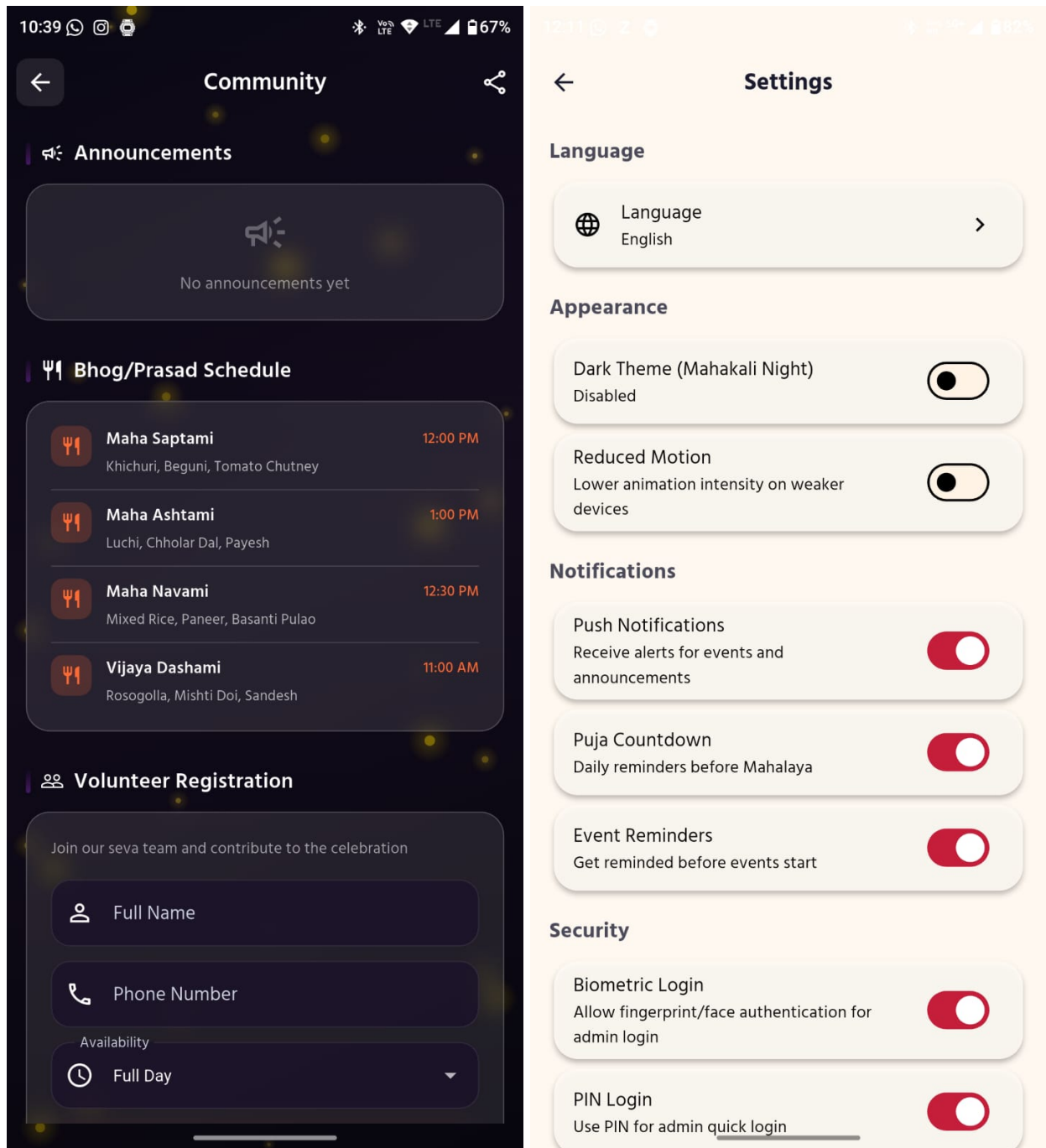


Figure 6: Real-time community feed and live theme switcher (Light/Dark/Festive modes).

5 Design Process & UI/UX Highlights

- **Research:** 3-day user interviews + competitive analysis of existing Puja apps.
- **Wireframing:** Low-fidelity in Figma → high-fidelity prototypes with 12 micro-interactions.
- **Color Theory:** Primary palette derived from traditional Durga Puja — #B71C1C (deep red), #FFD700 (gold), #0D47A1 (deep blue).
- **Typography:** Playfair Display (headings) + Poppins (body) for perfect readability.
- **Motion Design:** 8 custom animations using Flutter's AnimationController; Goody effect via

CustomPainter + bezier curves.

All decisions were validated through 5 usability testing rounds with target users.

6 Challenges Faced & Solutions

- Challenge: Creating liquid Goody effect in Flutter → Solution: CustomPainter with animated bezier paths.
- Challenge: Offline-first gallery with large images → Solution: CachedNetworkImage + asset fallbacks.
- Challenge: Theme switching without reload → Solution: Provider + AnimatedTheme widget.

7 Conclusion & Future Roadmap

Impact: This app has the potential to digitize and enhance Durga Puja celebrations for 10,000+ users in its first year.

Future Improvements (Post-Hackathon)

- Real Razorpay/UPI integration
- AR/VR pandal preview using ARCore
- Admin panel for organizers (Flutter Web)
- AI-powered event recommendations
- Multi-city expansion + 10+ regional languages
- Blockchain-based transparent donation ledger
- Integration with Google Calendar & Maps API

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- GDG On Campus SIESGST and Friends of Figma Mumbai for organizing PixelVerse 2026
- Flutter community and official documentation
- Durga Puja committees who shared cultural insights
- Our mentors and family for continuous support

This project truly embodies “*Where Logic Meets Aesthetic*” — a functional masterpiece wrapped in cultural beauty.

We are excited to present this live during the offline round on **8 March 2026** at SIESGST, Nerul. Thank you for your valuable time and consideration!